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Millifluidic triMEA organoid platform with stretchable microelectrode arrays for parallel organoid electrophysiology under standardized perfusion

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Organoids enable rigorous studies of neurodevelopment and disease, yet existing platforms often compromise between precise microenvironment control and reliable electrophysiological measurement. The challenge of achieving uniform perfusion alongside multi-well volumetric neural recordings limits parallel functional characterization and technical reproducibility across samples. Here, we present the triMEA millifluidic platform, which integrates three stretchable microelectrode arrays within a single device and delivers uniform perfusion across three independent wells. A single inlet supplies identical perfusion to each well, which stabilizes flow patterns and equalizes mechanical stimuli between chambers. This design facilitates simultaneous, independent electrophysiological recordings from three organoids within a single experimental setup. Using human midbrain organoids as a proof-of-concept biological model, we provide computational and experimental validation of the fluidic profiles and demonstrate electrophysiological recordings across all three wells under continuous perfusion. This platform establishes a reproducible and standardized foundation for parallel organoid interfacing.

1 Introduction

Brain organoids are increasingly used as platforms for studies of neurodevelopment and neurological disorders due to their ability to recreate human neuronal network architecture and functional responses *in vitro*.[1–3] To sustain organoids for longitudinal characterization, controlled perfusion systems are essential. These systems must deliver defined biochemical cues and mechanical stimuli, promote self-organization and differentiation, and ensure efficient nutrient delivery, waste removal, and oxygen transport.[4,5] When these conditions are met, organoids progressively develop spontaneous and evoked electrical activity, with increasing firing rates and network synchrony as tissue matures.[6] By precisely controlling microenvironmental parameters and perfusion conditions, these platforms enable the engineering of disease-specific states that reproduce characteristic electrical activity patterns. This, in turn, provides robust electrophysiological readouts and supports the development of accurate disease models.[7]

To date, commercially available microelectrode arrays (MEAs) have been widely used for electrophysiological recordings, cell viability assessment, and tissue health monitoring.[8–13] Custom fabricated MEAs have also been developed to address diverse recording and stimulation requirements beyond those of commercial systems.[14–16] However, both remain limited to surface-level recordings that are unable to capture volumetric neural activity throughout an organoid. Furthermore, neither system incorporates integrated perfusion, which limits their utility for maintaining homeostasis and ensuring stable recording conditions over time. To overcome these limitations, researchers have developed systems in which microfluidic modules are mounted onto flat substrates with planar microelectrodes.[17,18] In these systems, perfusion is achieved by flowing medium through

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microfluidic channels that border or cover the organoid chamber, ensuring nutrient exchange, waste removal, and a stable biochemical environment. While these approaches reduce contamination risks, electrophysiological measurements remain confined to the organoid surface.

Unlike cardiac tissue, where activity propagates as a coordinated wavefront, brain organoids generate spontaneous spikes at spatially distinct and random locations throughout the tissue volume, which requires high spatial resolution in three dimensions to capture network-level activity.[11] To achieve more comprehensive recordings, conformal three-dimensional (3D) arrays that can wrap around or penetrate the organoid have been developed. Examples include self-actuating MEAs,[19,20] stretchable MEAs,[21–24] mesh MEAs,[25] and reshapable MEAs,[26] all designed to increase the spatial coverage for detecting spontaneous neural activity. Among these, stretchable MEAs accommodate a range of organoid sizes without the constraints of size-specific flexible designs[21] and maintain conformal contact with the organoid surface under continuous media flow and organoid movement.[24] This sustained contact is critical for a platform incorporating continuous perfusion, where fluid flow and organoid growth over time can otherwise disrupt electrode-tissue interfaces. These advances result in higher spike counts, larger amplitudes, and more complete network maps that capture spontaneous bursts and oscillations across the organoid volume. Despite these improvements, most current approaches are limited to a single well, which prevents the parallel assessment of multiple samples and limits the ability to ensure technical reproducibility across experiments. Instead, the current systems often rely on static culture or uncontrolled perfusion, which leads to environmental instability and inconsistent functional data. Although one study combined a 3D MEA with a perfusable microfluidic system,[22] it was still limited to a single well and did not enable parallel evaluation or validate technical consistency across organoids. Consequently, studies restricted to single-well formats require sequential experimentation on individual organoids, which slows data acquisition, limits sample size, and prevents the simultaneous assessment of functional activity across distinct organoid populations under identical conditions. Therefore, there is a need for platforms that enable parallel evaluation with integrated perfusion. Such systems are vital for establishing standardized recording conditions and ensuring that biological variability can be accurately distinguished from technical variations in the testing environment and interface.[12,27]

Here, we introduce the triMEA platform, a millifluidic organoid-on-a-chip that integrates three stretchable three-dimensional microelectrode arrays (sMEAs) within a single device designed for uniform parallel perfusion and simultaneous electrophysiological recording across three independent wells (Fig. 1). Finite element modeling (FEM) of the fluidic domain was developed to optimize the channel geometry, which resulted in uniform perfusion and matched shear stress across all three wells. The optimized design was subsequently validated through computational and experimental characterization. We assessed the functionality of our platform using midbrain organoids (mBOs) generated from human embryonic stem cells (hESCs). Baseline functional activity was monitored in all three wells under continuous perfusion, confirming that the platform maintains signal quality at all recording sites throughout active flow conditions. These combined features establish a technical foundation for parallelized functional characterization of brain organoids and for future longitudinal studies requiring standardized recording conditions.

2 Methods

2.1 Computational fluid dynamics

CFD simulations were performed in COMSOL Multiphysics, assuming steady-state incompressible Newtonian laminar flow (Navier–Stokes and continuity). A fully developed laminar inflow profile was prescribed at the inlet, and the volumetric flow rate was swept from 0.001 to 1.0 mL min^{-1}. The outlet boundary was set to 0 Pa gauge pressure, equivalent to atmospheric pressure at the outlet port. No-slip boundary conditions were applied to all channel walls and the organoid surface. The organoid was represented as a sphere centered in the chamber, with a density of 1041 kg m^{-3} and a Young's modulus of 9921 Pa. The sMEA structure was excluded from the geometry during the parametric optimization sweep. A separate set of simulations incorporating the sMEA structure was performed on the final optimized geometry.

Using the resulting steady-state velocity field, time-dependent solute transport was simulated using a convection–diffusion model. Concentration was set to 1 at the inlet and 0 as the initial condition throughout the domain. Simulations were run from 0 to 300 s for higher flow rates, with results recorded every 0.1 s, and from 0 to 3000 s for lower flow rates, with results recorded every 50 s. Channel and organoid surfaces were treated as impermeable, using a zero normal flux boundary condition. Flow uniformity was quantified as the surface-averaged wall shear stress on the organoid surface, and compound delivery was quantified as the volume-averaged concentration within each well. Uniformity across wells was summarized as the coefficient of variation of the well-averaged concentration.

2.2 Oxygen concentration modeling

A three-dimensional computational model was developed to simulate oxygen (O_2) transport and consumption within mBOs using the transport of diluted species module in COMSOL Multiphysics. The model incorporates both convective and diffusive transport processes coupled with nonlinear O_2 consumption kinetics to capture physiologically relevant O_2 distribution profiles and hypoxic core formation. Fick's law was used to describe O_2 diffusion within both media and organoids under the assumption of incompressible flow:

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$$\frac{\partial c}{\partial t} + \nabla \cdot J + u \cdot \nabla c = R$$

$$J = -D\nabla c$$

where c is the O_2 concentration (mol m^{-3}), D is the diffusion coefficient, and u is the velocity field (m s^{-1}). Diffusion coefficients were set to 3×10^{-9} m^2 s^{-1} in the media and 1×10^{-9} m^2 s^{-1} within the organoid.[28] The operator ∇ denotes the standard vector differential operator,

$$\nabla = \hat{i}\frac{\partial}{\partial x} + \hat{j}\frac{\partial}{\partial y} + \hat{k}\frac{\partial}{\partial z}$$

where $u \cdot \nabla c$ represents advective flux, and $\nabla \cdot J$ represents diffusive flux. The outer boundary of the media domain was maintained at an O_2 concentration of 0.2 mol m^{-3}, corresponding to the equilibrium concentration under atmospheric conditions.

Within the organoid, O_2 consumption by cells was modeled using Michaelis–Menten kinetics combined with a smoothed switching function to reproduce hypoxic shutdown of metabolic activity. The volumetric reaction rate was defined as:

$$R = -V_{max}\left[\frac{c}{c + k_M}\right] \cdot \delta(c > c_{crit})$$

Here, V_{max} is the maximum O_2 consumption rate (mol m^{-3} s^{-1}), k_M denotes the Michaelis–Menten constant, representing the concentration at which consumption reaches half of V_{max}. In this model, k_M was set to 0.005 mol m^{-3}.[29] The critical concentration c_{crit}, below which metabolic consumption ceases due to hypoxia-induced necrosis, was set to 0.05 mol m^{-3}.[30] To create a smooth step down once the critical concentration is reached, we used the Heaviside function with a continuous second derivative (flc2hs) function to define δ. The initial and boundary media concentration c_0 was set to 0.2 mol m^{-3}, corresponding to the equilibrium concentration under atmospheric conditions.[31] All simulations were conducted at an incubation temperature of 37 °C.

2.3 Fabrication of stretchable microelectrode array

First, a sacrificial layer of 300 nm of poly(methyl methacrylate) (PMMA) (950 PMMA A4, Kayaku Advanced Materials Inc., Japan) was spin-coated on a four-inch silicon wafer and baked at 180 °C for 90 s (Fig. S1). Next, we spin-coated a 4 μm-thick polyimide (PI) (GPI-100A, PNS Technology, Korea) layer and fully imidized it at 80 °C, 150 °C, 230 °C, and 280 °C for 30 min at each step. Then, we patterned a 2 μm-thick negative photoresist (PR) (NR9-3000PY, Futurrex Inc., USA) on top of the PI layer, followed by the deposition of titanium (Ti) and gold (Au) (30/200 nm) by electron beam evaporation. To form the microelectrodes and signal lines, the Ti/Au layer was lifted off. For device passivation, a 1 μm-thick PI layer was spin-coated and cured under the same conditions described above. For device body patterning and the opening of microelectrodes and contact pads, a 50 nm-thick copper (Cu) hard mask was patterned, and the PI layer was subsequently etched using O_2 plasma (300 W, 50 sccm). The Cu hard mask was removed by immersing the sample in APAL-1 (APCT, Korea) for 40 s.

2.4 Electrodeposition of conductive PEDOT : PSS layer

Before the release of the sMEA from the Si substrate, electrodeposition of PEDOT : PSS was conducted to form protruding microelectrodes. The device on Si was interconnected with a flexible printed circuit board (fPCB) using an anisotropic conductive film (ACF) bonder (OL-FD006, Shenzhen Olian Automatic Equipment, China). To electrochemically activate the microelectrodes prior to PEDOT : PSS deposition, cyclic voltammetry (CV) was performed in 0.5 M H_2SO_4 using a voltage sweep from −0.5 V to 1.5 V, repeated 5 times. Next, a mixture of 0.5 M poly(sodium 4-styrenesulfonate) (PSS) (243051, Sigma-Aldrich, USA) and 0.05 M 3,4-ethylenedioxythiophene (EDOT) (483028, Sigma-Aldrich, USA) was added to deionized water (DI) and sonicated at high power for 30 min. Following sonication, 50 μL of the mixed PEDOT : PSS solution was deposited on the sMEA surface. Electrodeposition was performed on individual microelectrodes using a two-electrode configuration, with the device connected to a potentiostat (WizEIS-1200 Premium, WizMAC, Korea) and a platinum (Pt) wire used as the counter electrode. A current of 300 nA was applied for 525 s per electrode. After the process, the sMEA was rinsed with ethanol and deionized water.

2.5 Millifluidic device fabrication

Millifluidic molds incorporating three well openings, inlet/outlet channels, and a groove for the fPCB connector were designed in COMSOL and fabricated using a high-resolution stereolithographic 3D printer (Form 3, Formlabs, Massachusetts, USA) with standard white resin at 50 μm resolution (Fig. S2). Subsequently, the printed parts were stirred in isopropyl alcohol (IPA) for 90 min using an agitator (Form Wash, Formlabs, Massachusetts, USA), air-dried under ambient conditions for 6 h, and post-cured under UV light at 60 °C for 60 min (Form Cure, Formlabs, Massachusetts, USA). To form the PDMS casting chamber, three printed sidewalls were manually attached to the primary mold structure, creating an open-top enclosure. Each sidewall was made with an internal groove to guide and support a glass slide, while the rear wall served as a physical stop. A standard glass slide (76 × 52 mm) was treated with 1.0 μL of trichloro(1*H*,1*H*,2*H*,2*H*-perfluorooctyl) silane (Sigma-Aldrich, Korea) *via* vacuum desiccation for 2 min to prevent adhesion. The 3D-printed molds were treated with ER-200 (Mann Release Technologies Inc., USA) to facilitate PDMS detachment after curing. The slide was inserted along the grooves until it contacted the rear wall, closing the bottom of the chamber, which enabled elastomer casting. PDMS (Sylgard 184, Dow Corning Co., USA) was prepared by mixing the base and curing agent in a 10 : 1 ratio, followed by degassing for 30 min. The prepolymer was then poured into the mold through the top opening, degassed again, and cured at 80 °C

for 4 h in a vertical orientation in a vacuum oven. After curing, the side and rear walls were disassembled, allowing the glass slide protruding beyond the PDMS layer to be pushed out from below. The PDMS layer was then demolded, and a mixture of PDMS base, curing agent, and hexane (Sigma-Aldrich, Korea) at a ratio of 50 : 5 : 1 was applied *via* syringe to areas contacting the mold, excluding the channel and fPCB interfaces, followed by spin coating to dissolve contaminants and improve PDMS–PDMS bonding. Then, the layers were cured in the oven at 80 °C for 4 h, and inlet/outlet ports (Ø 2 mm) were punched using a biopsy punch (Miltex Instrument, USA). Prior to bonding, the sMEA was released from the silicon substrate by immersion in acetone for 6 h to dissolve the sacrificial PMMA layer. A water-soluble polyvinyl alcohol (PVA) tape was applied to the released sMEA to support the serpentine structures during handling. The released sMEA was then inserted into the fPCB connector seated within the corresponding groove in the PDMS mold, which set the sMEA position within the well without a separate alignment step. The PVA tape was subsequently dissolved in warm phosphate-buffered saline (PBS) once the sMEA was secured in place. The PDMS layers were temporarily protected using adhesive tape (Scotch, 3 M, USA) and subsequently bonded after exposure to O_2 plasma in reactive ion etching (RIE) at 20 W and 20 sccm for 5 s. The assembled layers were bonded to a glass substrate under O_2 plasma for a second time (20 W, 20 sccm, 5 s). Finally, the chip chambers were sealed by applying a MicroAmp™ optical adhesive film (Thermo Fisher Scientific, Waltham, Massachusetts, USA) to the top surface of the assembled device, providing an optically transparent, resealable cap that maintains the pressure required for consistent medium flow while allowing organoid exchange between experiments (Fig. S3).

2.6 Electrochemical characterization

Electrochemical impedance spectroscopy (EIS) and CV were performed to measure electrode impedance in PBS using a potentiostat (Autolab PGSTAT204, Metrohm AG, Switzerland). A two-electrode configuration was used, with platinum (Pt) serving as a counter electrode. For EIS, a sinusoidal voltage signal with an amplitude of 50 mV was applied to the microelectrodes, and the frequency was swept logarithmically from 10 Hz to 1 MHz. For CV, a voltage sweep from −0.6 V to +0.8 V was applied at a scan rate of 0.05 V s^{-1}.

2.7 *In vitro* electrophysiological recordings

Prior to recording, the organoid platform was first sterilized by immersion in 70% ethanol for at least 24 h. After sterilization, the platform and channels were rinsed with sterile PBS, and a 1% Matrigel (Corning, USA) solution diluted in PBS was perfused through the platform. The sMEAs were stored in the diluted Matrigel solution at 4 °C for at least 8 h. Subsequently, culture medium suitable for the cell type was perfused inside the platform for at least 30 min to remove residual Matrigel. The platform was then placed in an incubator (NB203, N-BIOTEK, Korea) for 1 h to equilibrate the medium temperature to 37 °C. For electrophysiological recording, the organoids were transferred from their respective culture wells to the platform. Following transfer, the organoids were allowed to stabilize on the sMEAs for 30 min prior to signal acquisition. Upon completion of the recording, the organoids were removed, and the sMEAs were prepared for reuse by replacing the culture medium with trypsin (Gibco, USA), then incubated at 37 °C for 12 h to remove residual tissue. The device was then sterilized in 70% ethanol and recoated with Matrigel to restore biocompatibility.

To perform recording, the sMEAs inside the incubator were connected to an external amplifier board (RHD2000, Intan Tech., USA) *via* a wire grommet. The recording bandwidth was set to 1–7500 Hz, with a sampling rate of 30 kHz. Action potentials were processed using a custom spike sorting app developed in MATLAB (MathWorks, USA). Prior to sorting, the measured signals were filtered using a bandpass filter (300–6000 Hz). Spikes were detected for each channel using a threshold-based peak detection algorithm. The noise floor was estimated using the median absolute deviation (MAD) of the bandpass-filtered signal calculated as σ_n = median ($|S(t)|$)/0.6745, where $|S(t)|$ is defined as the absolute amplitude of the bandpass-filtered signal. The detection threshold was set to be 8 σ_n. Peaks occurring within 1 ms of one another were excluded to avoid redundant detection, as were simultaneous peaks detected across multiple electrodes. Spikes were segmented into 10 ms windows (5 ms before and 5 ms after the threshold crossing), and the corresponding timestamps were used to generate the raster plot. Differences in recording noise between static and perfused conditions were assessed using a two-sided Mann–Whitney *U* test, with Bonferroni correction applied across the three positional comparisons (left, center, right). Statistical significance was defined as corrected $p < 0.05$.

2.8 Generation of midbrain organoids

This study was conducted in accordance with relevant ethical guidelines and approved by the Public Institutional Bioethics Committee of the Ministry of Health and Welfare (MoHW), Republic of Korea (IRB no. P01-201409-ES-01), for experiments involving hESCs. To generate mBOs, hESCs were dissociated with Accutase (Merck, Germany), and 8×10^3 cells were seeded in a 96-well low-attachment plate containing mTeSR™1 basal medium (STEMCELL Technologies, Canada) with 50 μM Y-27632 (Tocris, United Kingdom). From day 0, after the cells formed embryoid bodies (EB), the medium was exchanged with N2 base medium [1 : 1 mixture of DMEM/F-12 (1×) (Gibco) and Neurobasal™-A medium (Gibco), 1× N2 supplement (Gibco), 1× GlutaMAX supplement (Gibco), 20 U mL^{-1} penicillin/streptomycin (Gibco), 1× MEM non-essential amino acids (Gibco), and 50 μM 2-mercaptoethanol (Gibco)] containing 10 μM SB431542 (Tocris), 150 ng mL^{-1} Noggin (R&D Systems, Minnesota, USA), 400 ng mL^{-1} Shh-C24II (Miltenyi Biotec, Germany), and 1.5 μM CHIR99021 (Tocris) for neural induction. The N2 medium containing the above supplements was replaced on days 2, 4, and 6. On day 8, the medium was replaced with N2 medium supplemented with 100 ng mL^{-1} FGF8b (PeproTech) for

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midbrain patterning. On day 11, the medium was exchanged with B27 base medium [Neurobasal™ – A medium supplemented with 1× B27 supplement without vitamin A (Gibco), 1× GlutaMAX supplement (Gibco), 20 U mL^{-1} penicillin/streptomycin (Gibco), 1× MEM non-essential amino acids (Gibco), and 50 μM 2-mercaptoethanol (Gibco)] containing 100 ng mL^{-1} FGF8b, 20 ng mL^{-1} BDNF (PeproTech), and 200 μM ascorbic acid (Sigma-Aldrich, St. Louis, Missouri, USA). On day 14, organoids were matured and maintained in B27 base medium supplemented with 20 ng mL^{-1} BDNF, 10 ng mL^{-1} GDNF (PeproTech), 200 μM ascorbic acid, 500 μM cAMP (Biogems), and 1 μM DAPT (Sigma-Aldrich, Missouri, USA) for terminal differentiation. The maintaining medium was changed every other day.

2.9 Quantitative real-time PCR (qRT-PCR)

To test mRNA expressed in mBOs, total RNA was extracted from mBOs using Easy-BLUE™ Total RNA extraction kit (iNtRON, Korea), following the manufacturer's protocol. A total of 1200 ng RNA was reverse transcribed into cDNA using SuperScript IV VILO (Thermo Fisher Scientific). Quantitative PCR was performed using Fast SYBR™ Green PCR Master Mix (Applied Biosystems, Waltham, Massachusetts, USA) *via* the QuantStudio™ 5 Real-Time PCR System (Applied Biosystems) with gene-specific primers as follows: *SOX2*: forward 5′-AGAACC CCAAGATGCACAAC-3′ and reverse 5′-ATGTAGGTCTGCGAGC TGGT-3′; *PAX3*: forward 5′-TTTCCGTTTCGCCTTCACCT-3′ and reverse 5′-ACGATCTTGTGGCGGATGTG-3′; *LMX1a*: forward 5′-AAAGCGCGATCGACACCTC-3′ and reverse 5′-TCCCGGTAGAA GCAGGTGGT-3′; *MAP2*: forward 5′-CCGTGTGGACCATGGG GCTG-3′ and reverse 5′-GTCGTCGGGGTGATGCCACG-3′; *TH*: forward 5′-GGGCGTTGTAAGCAGAACG-3′ and reverse 5′-AAGG CCCGAATCTCAGGCT-3′; and *GAPDH*: forward 5′-GAAGGTGAA GGTCGGAGTC-3′ and reverse 5′-GAAGATGGTGATGGGATTTC-3′. The comparative Ct method was used, and relative mRNA expression was calculated based on normalizing to *GAPDH*.

2.10 Immunostaining

The mBOs were fixed with 4% paraformaldehyde (Biosesang, Korea) and dehydrated in 30% sucrose in PBS. The organoids were embedded in OCT compound (Sakura, Japan), frozen, and sectioned from the frozen block on a cryostat (CM1520, Leica Biosystems, Germany). The sectioned samples were permeabilized for 1 h with 0.3% Triton X-100 (Sigma-Aldrich, USA) in TBS-T (0.025% Tween-20 in TBS, Sigma-Aldrich, USA) and then blocked with 3% BSA (Thermo Scientific) in TBS-T for 1 h at room temperature. The samples were incubated with 1 : 200 diluted primary antibody against SOX2 (Invitrogen, Waltham, Massachusetts, USA, #PA1-094), MAP2 (Sigma-Aldrich, #MAB3418), TH (Novus Biologicals, Littleton, Colorado, USA, #NB300-109), NESTIN (Abcam, Cambridge, United Kingdom, #ab6320), GFAP (Dako, #Z0334) at 4 °C overnight, and then stained with 1 : 1000 diluted secondary antibodies (Alexa Fluor 488-conjugated goat anti-mouse IgG (Life Technologies) and Cy3-conjugated anti-rabbit IgG (Life Technologies)) for 1 h at room temperature. Nuclei were counterstained with Hoechst 33258 (Invitrogen) for 5 min. The stained samples were mounted with fluorescence mounting medium (Dako, Denmark) and observed using a confocal microscope (LSM800, ZEISS, Germany).

3 Results and discussion

3.1 Overview of millifluidic triMEA platform

Developing a multi-well platform for parallel organoid analysis requires solving two key engineering challenges: achieving uniform fluid perfusion across all wells and ensuring that continuous flow does not degrade electrophysiological signal quality. To address these requirements, we designed the triMEA platform to provide three organoids with identical mechanical, biochemical, and electrical interfacing conditions. The platform comprised three PDMS layers (bottom, middle, and top) that formed millifluidic channels and three cylindrical organoid chambers (Fig. 1a and b). The top layer was sealed with optical adhesive, which maintained the pressure required for consistent medium flow and allowed organoid exchange between experiments, and the bottom layer was bonded to a glass substrate (Fig. 1b). Each sMEA comprised an array of stretchable gold electrodes coated with PEDOT : PSS, which increased electrode surface area (Fig. 1b, inset). Each organoid was placed on a dedicated sMEA at the chamber center, with medium directed across the organoid surface within the well (Fig. 1c). The fabricated device featured a shared inlet that distributed medium through three parallel channels into each well, and a shared outlet that collected medium through three parallel channels from each well (Fig. 1d). The PEDOT : PSS was over-electrodeposited into protruding electrode structures that formed a tight contact with the organoid surface, which enabled extracellular signal detection under continuous perfusion (Fig. 1d, inset). Each sMEA was electrically interfaced with the fPCB *via* anisotropic conductive film (ACF) bonding, which provided electrical access to the external recording system and accommodated the inlet and outlet tubing without interference. The platform delivered uniform flow distribution across all three wells (Fig. 1e).

3.2 Computational modeling of flow dynamics

To ensure that observed differences in neural activity reflect biological responses rather than environmental artifacts, it is critical to minimize technical variability across recording sites. This requirement necessitates consistent fluid flow and safe mechanical conditions across all wells. We utilized shear stress as our primary evaluation metric, as it provides a direct measure of local mechanical conditions and serves as a computationally accessible proxy for flow velocity distribution.[32–34] Shear stress experienced by the organoid is defined as:

$$\tau = \mu \frac{\partial v}{\partial y}$$

where τ is the shear stress at the organoid surface, μ is the dynamic viscosity, and $\partial v/\partial y$ is the velocity gradient normal to the organoid surface. Assuming constant viscosity (μ) and

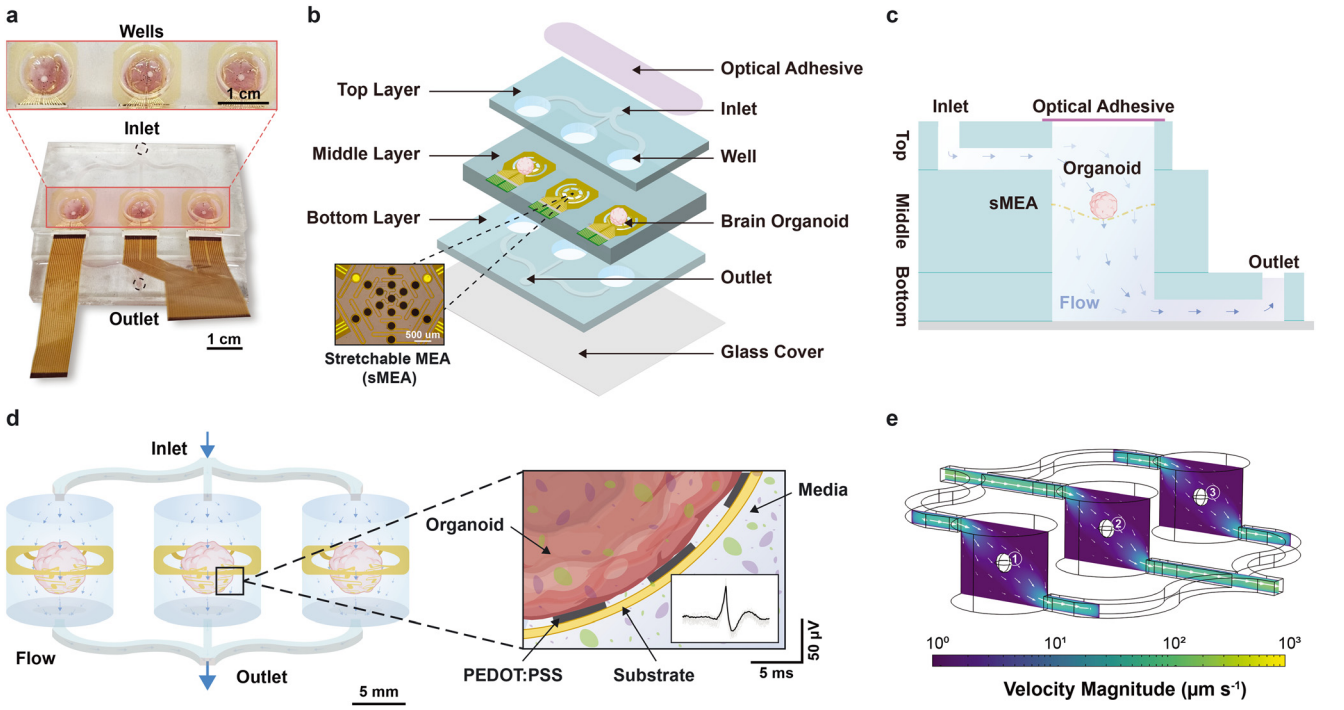


Fig. 1 Millifluidic triMEA platform for organoid monitoring. a) Optical image of the packaged triMEA platform. b) Exploded view of the triMEA platform. Inset shows the layout of the sMEA microelectrodes. c) Cross-sectional view of the platform. d) Schematic of the three-well flow architecture, with a zoomed inset depicting the sMEA–organoid interface (Au and PEDOT : PSS layers) and a representative spike recorded during perfusion. e) Representative simulation of flow velocity magnitude across the three wells.

identical geometries, uniform shear stress guarantees consistent flow rates and equal convective transport of compounds across all wells. Furthermore, we ensured that this shear stress remained within physiologically tolerable ranges (Table S1). While excessive mechanical stress (>100 mPa) can damage tissue integrity,[35,36] moderate shear promotes nutrient exchange without tissue damage.[35–39] Therefore, our design objective was to maintain a uniform shear stress below 1 mPa throughout the platform to prevent tissue damage.

While scaling to a three-well platform effectively triples data output by enabling synchronous multi-site electrophysiology, achieving uniform flow distribution across these multiple wells proved challenging.[40–42] Initial series-connected designs and simple parallel layouts resulted in unequal and inconsistent flow distributions (Fig. S4). To overcome these geometric limitations, we employed FEM to systematically optimize the channel architecture. We modeled laminar flow and species transport in COMSOL under Newtonian, incompressible, and no-slip conditions. The simulation geometry consisted of the full three-well device, including the inlet, three branch channels, three wells, and outlet. Within each well, the organoid was represented as a sphere centered at the well. This central position approximated the typical organoid location on the sMEA. Although actual organoids exhibit more complex morphologies, this spherical approximation effectively captures key flow effects. While localized deviations, such as surface irregularities and gravitational flattening, may slightly alter local shear stress, the bulk flow distribution across the wells is primarily governed by the channel geometry and remains robust against these minor shape variations.

3.3 Parametric optimization of channel geometries

We systematically evaluated five geometric parameters: center channel width (w_c), channel height (h), inlet/outlet distance to well (L_v), and two side channel width parameters. Each side channel consists of a longer segment that spans most of the channel path (w_{s1}) and a shorter segment immediately adjacent to the well (w_{s2}). Width values are applied symmetrically across the left and right side channels and across the corresponding inlet-side and outlet-side channels to balance flow distribution across the wells (Fig. 2a and Table 1). Because the central fluidic path is naturally shorter, flow balance was achieved primarily by constricting the center width (w_c) to increase hydraulic resistance according to:

$$R_h = \frac{12\mu L}{wh^3}$$

where μ is dynamic viscosity, L is channel length, w is width, and h is height. Systematic parameter sweeps confirmed that this center constriction (w_c) was the dominant factor controlling flow distribution (Fig. 2c), while the inlet/outlet distance to well (L_v) provided secondary fine-tuning (Fig. 2d). An increase in the side channel width, section 1 (w_{s1}), corrected an initial imbalance in which the center well exhibited higher shear stress

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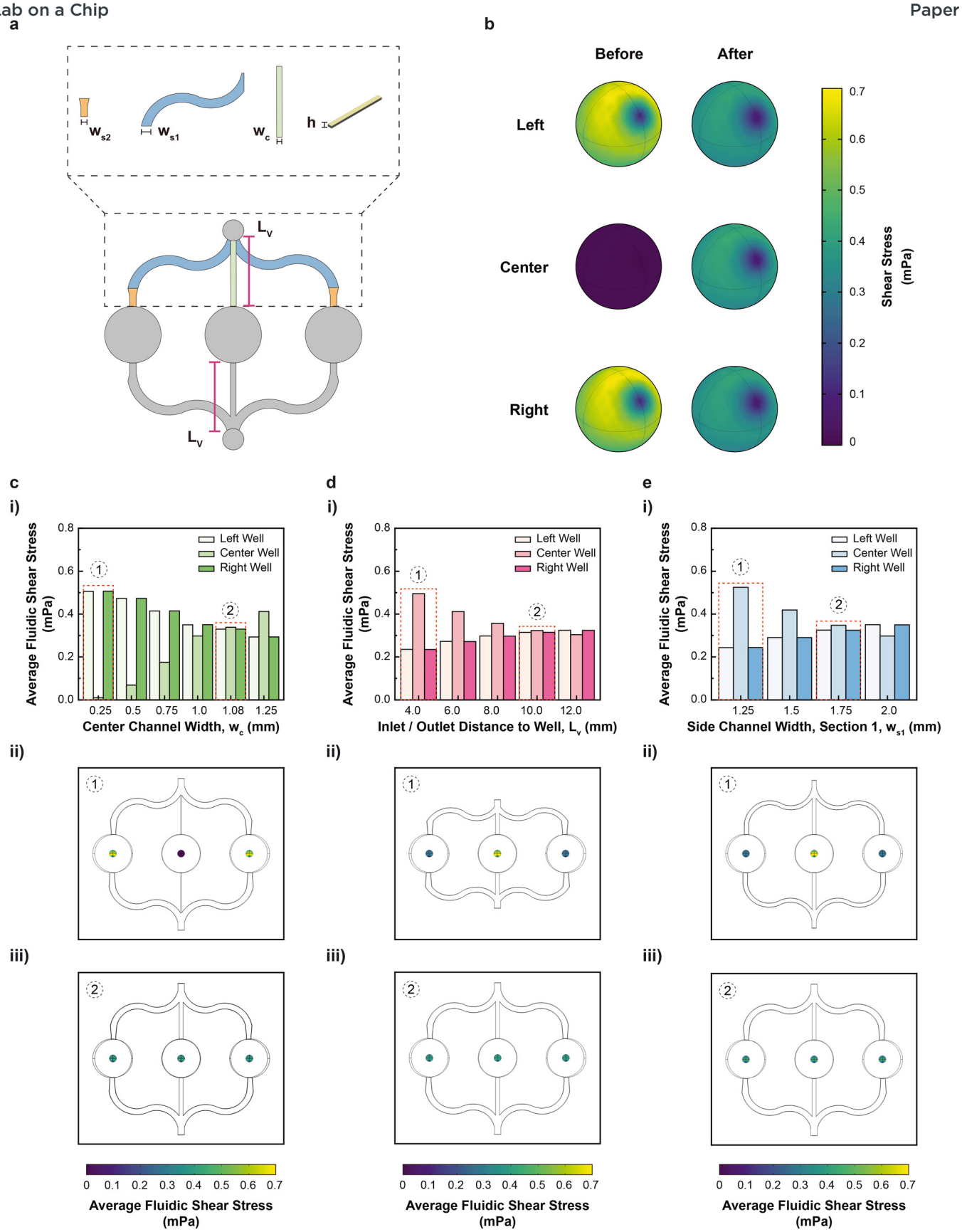


Fig. 2 Geometric optimization of the triMEA platform for uniform multi-well perfusion. a) Schematic illustration of channel geometry with design parameters. b) Surface shear stress distributions on left, center, and right organoids before and after geometric optimization. The "before" condition corresponds to an initial, non-optimized geometry with all channel dimensions set to the lower bound of their tested range. c–e) Effects of c) center channel constriction width (w_c), d) inlet/outlet distance to well (L_v), and e) side channel width, section 1 (w_{s1}) on average fluidic shear stress in each well. For each parameter, i) average fluidic shear stress per well across the tested range, with dashed lines at the (1) initial (suboptimal) and (2) final (optimized) values, ii) shear stress distribution at (1), and iii) shear stress distribution at (2).

Table 1 Simulation parameter ranges

Parameters	Simulation range (mm)
Side channel width, section 1 (w_{s1})	1.25–2.00
Side channel width, section 2 (w_{s2})	1.00–1.75
Center channel width (w_c)	0.25–1.25
Channel height (h)	0.50–2.00
Inlet/outlet distance to well (L_v)	4.00–12.00

than the left and right wells, and equalized shear stress across the three wells (Fig. 2e). The sweep used 0.25 mm intervals, and the final value was refined between grid points. The side channel width, section 2 (w_{s2}), and channel height (h) showed minimal influence on overall distribution, but were optimized to ensure a gradual geometric transition that reduces flow separation (Fig. S5). Additional analyses confirmed that this optimized flow distribution was robust against variations in well geometry, organoid size, and the presence of sMEA (Fig. S6). Under the final optimized parameter set (w_c = 1.08 mm, w_{s1} = 1.89 mm, w_{s2} = 1.24 mm, L_v = 10 mm, and h = 2.0 mm), we achieved <3% well-to-well variation at 0.1 mL min^{-1}. Follow-up simulations incorporating the sMEA structure were performed on this optimized geometry and demonstrated uniform shear stress across the bulk organoid surface, with values safely below the damage threshold and localized zero-shear regions limited to the electrode contact points (Fig. S7).

Beyond flow uniformity, we evaluated oxygen delivery as a secondary benefit. Although organoids were cultured off-chip and transferred to the platform only for recording, we modeled oxygen transport to ensure adequate oxygenation during acute sessions. Intraorganoid oxygen concentration was calculated using a modified Michaelis–Menten consumption term. Under 0.1 mL min^{-1} perfusion, the predicted intraorganoid oxygen level at the organoid surface increased by approximately 57% to 0.112 mol m^{-3} compared with the static condition, and the thickness of the region above the critical viability threshold (c_{crit} = 0.05 mol m^{-3}, below which hypoxic stress and loss of function occur[30]) increased by approximately 18% (Fig. S8). These results show that convective oxygen delivery through perfusion improves oxygenation during acute recording, which reduces the risk of hypoxic stress.[43–45]

3.4 Characterization of flow dynamics

Computational fluid dynamics simulations were performed across flow rates from 0.001 to 1.0 mL min^{-1} to evaluate shear stress distribution at each recording site (Fig. 3a). This sweep extended beyond the platform's intended operating range of 0.1 to 1.0 mL min^{-1} to confirm that shear stress uniformity was preserved beyond normal operating conditions. Across this range, shear stress scaled proportionally with flow rate, with all three wells exhibiting matched profiles and uniform distribution. At the lower operating bound (0.1 mL min^{-1}), shear stress ranged from 0.046 to 0.047 mPa across the wells, while at the upper operating bound (1.0 mL min^{-1}), values remained below the damage threshold, with less than 3% well-to-well variation. This characterization established that the optimized geometry preserves uniformity across a 1000-fold range in flow rate.

To validate the temporal uniformity of compound delivery at the lower operating bound, we simulated tracer species transport at 0.1 mL min^{-1}, representing slower delivery kinetics that might be preferred for extended exposure studies. Unlike static culture, where compounds reach each well by diffusion alone and can exhibit significant well-to-well timing differences, perfusion-driven delivery ensures synchronized exposure even at low flow rates. All three wells reached 90% of inlet concentration within approximately 3000 s (~50 min), with well-to-well concentration variation below 1% throughout the delivery profile and 0.02% at steady-state (Fig. 3b). The coefficient of variation exceeded 5% only during the first 13 min, where the low mean concentration inflated the ratio, and remained below 5% thereafter (Fig. 3c). Temporal matching was therefore maintained under slow perfusion conditions. While the equilibration time is longer than at higher flow rates, the sub-1% variation throughout suggests that all organoids experience similar concentration profiles regardless of the selected flow rate.

To experimentally validate these computational predictions, we performed blue dye perfusion experiments at 1.0 mL min^{-1}, the upper bound of the operating range, to enable faster compound delivery for acute screening applications.[46,47] Time-lapse imaging captured dye transport in two independent devices (Fig. 3d). The experimental dye propagation followed the same temporal profile as the computational prediction, and all three wells filled at synchronized rates. Quantitative analysis of dye intensity over 120 s revealed highly uniform delivery across devices, with high device-to-device reproducibility across both devices (Fig. 3e). The coefficient of variation across wells remained below 15% at all time points, with an average of 7.6%. The deviation from the computationally predicted sub-5% variation can be attributed to fabrication tolerances and the practical limits of intensity quantification of optical dyes. The observed experimental uniformity nevertheless confirms that the optimized geometry achieves functionally matched compound delivery across wells under operating conditions. Experimental dye validation was performed at 1.0 mL min^{-1} as the lower operating rate of 0.1 mL min^{-1} produced delivery kinetics too slow for time-lapse imaging. Uniform well-to-well distribution was observed at both flow rates, and delivery consistency was preserved across the full operating range.

3.5 Characterization of midbrain organoids

To enable electrophysiological recordings on the millifluidic triMEA organoid platform, we first generated mBOs from hESCs. Differentiation was performed using a stepwise protocol consisting of EB formation, neural induction, midbrain patterning, and maturation (Fig. 4a). After the 14 day patterning phase, organoids were maintained in maturation medium and transferred onto the millifluidic triMEA organoid platform for individual electrophysiological recording. Phase-contrast images from day 0 to day 24 showed progressive growth of the

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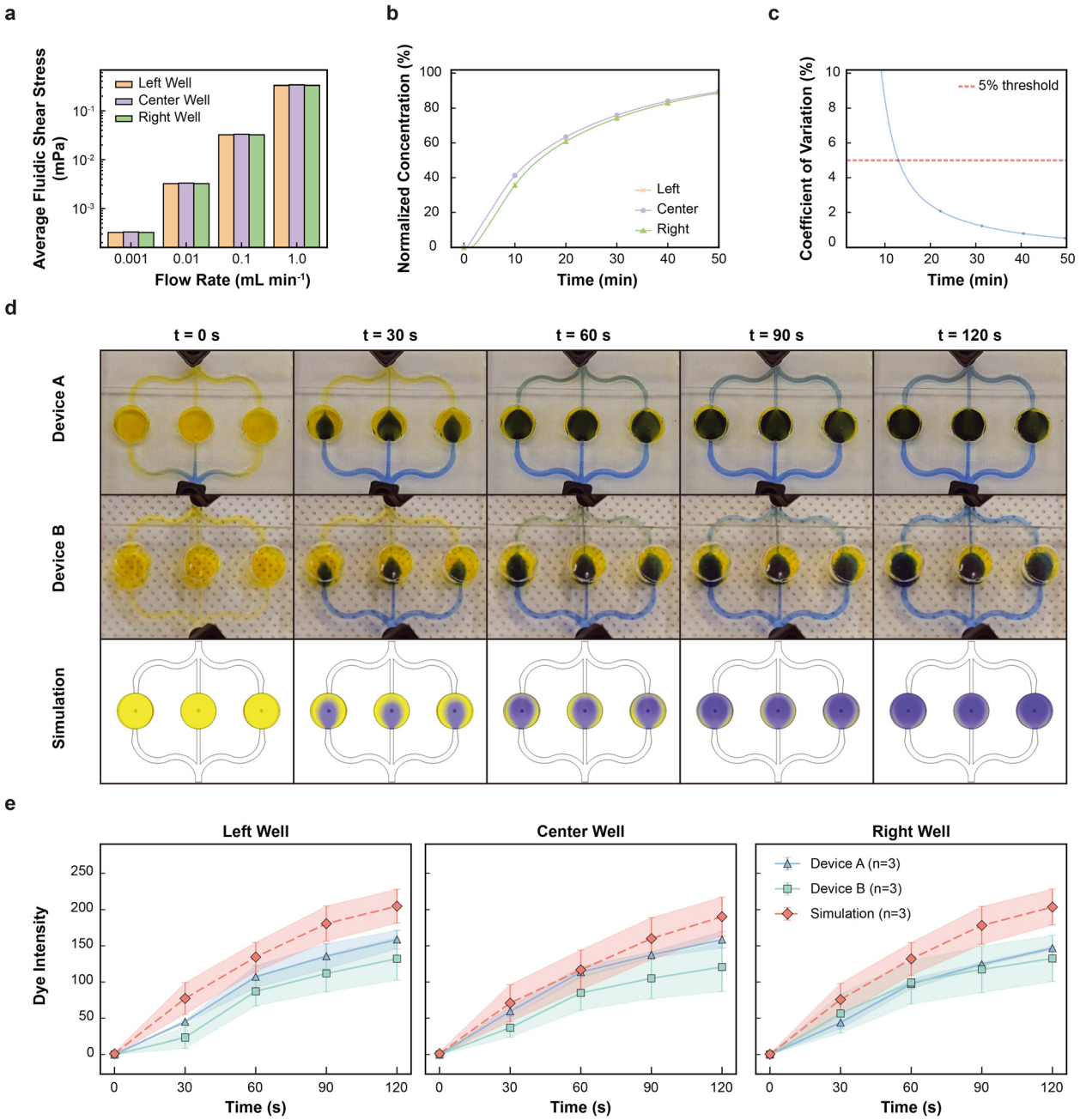


Fig. 3 Computational predictions and experimental validation of uniform flow distribution in the triMEA platform. a) Flow rate-dependent shear stress distribution across left, center, and right wells, across flow rates from 0.001 to 1.0 mL min^{-1}. b) Simulated normalized concentration profiles for tracer species at 0.1 mL min^{-1}. c) Coefficient of variation across wells over time. d) Experimental validation using blue dye perfusion at 0, 30, 60, 90, and 120 s in two independent devices (device A, device B) with a flow rate of 1.0 mL min^{-1}. The bottom row shows the corresponding COMSOL simulations. e) Quantification of dye intensity over time in left, center, and right wells for both devices and simulation (n = 3 independent repeats per condition).

organoids. To confirm mBO specification, we analyzed the mRNA expression of lineage-specific markers *via* qRT-PCR (Fig. 4b). The data showed that *SOX2* expression remained comparable to that of hESCs, suggesting the preservation of a neural progenitor pool within the organoids. Midbrain patterning was supported by the marked upregulation of *PAX3* and *LMX1a*, both associated with dopaminergic progenitor identity. In addition, the robust expression of *MAP2*, a mature neuronal marker, and *TH*, a key enzyme in dopamine synthesis, indicated the successful differentiation toward a dopaminergic neuronal lineage.

We next characterized the cellular composition of day 24 mBOs by immunostaining (Fig. 4c). The organoids exhibited a distinct spatial organization of various cell types. SOX2^{+} neural progenitors and NESTIN^{+} neural stem cells were present, indicating continued neurogenic potential. Extensive

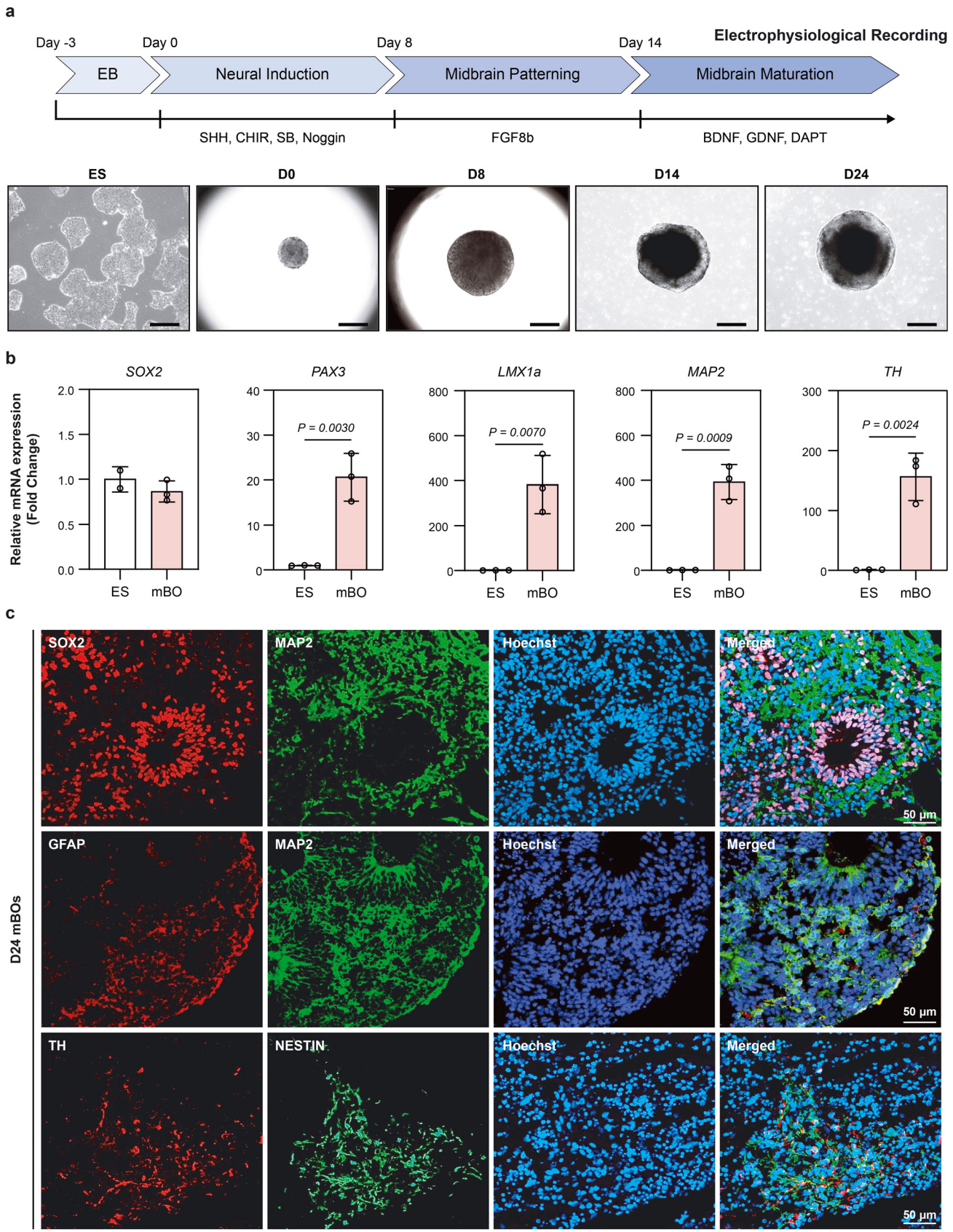


Fig. 4 Generation and characterization of mBOs differentiated from hESCs. a) Schematic overview of the differentiation protocol from hESCs to mBOs, with representative phase-contrast images showing organoid morphological progression on days 0, 8, 14, and 24. Scale bar = 400 µm. b) Relative mRNA expression levels of differentiation markers (*SOX2*, *PAX3*, *LMX1a*, *MAP2*, and *TH*) in undifferentiated hESCs and day 24 mBOs. Expression levels were normalized to *GAPDH*. Data are presented as the mean ± SD of independent experiments (n = 3). Statistical analysis was performed using an unpaired t test. c) Representative immunostaining images showing expression of SOX2, MAP2, GFAP, TH, and NESTIN in day 24 mBOs. Nuclei were counterstained with Hoechst 33258. Scale bar = 50 µm.

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$MAP2^+$ staining throughout the mBOs indicated a robust neuronal maturation into a post-mitotic neuronal state. The appearance of TH^+ cells confirmed the emergence of dopaminergic neurons, consistent with midbrain regional identity. $GFAP^+$ astrocytes were also detected, indicating the presence of glial support cells that contribute to neuronal maturation. Together, these results demonstrate that mBOs recapitulate key features of midbrain tissue, including appropriate cellular diversity and regional specification, and provide a suitable biological substrate for electrophysiological

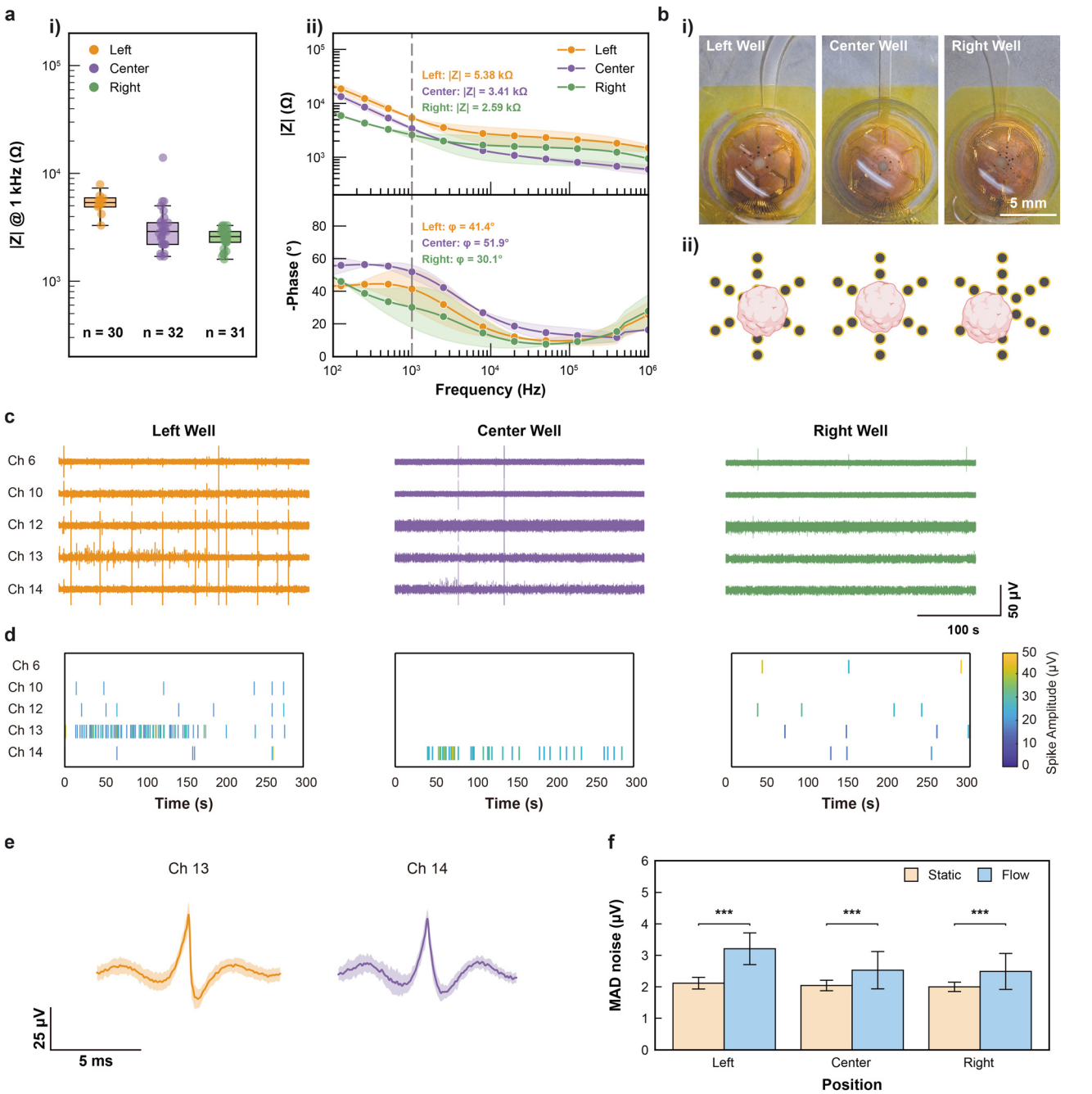


Fig. 5 Electrophysiological recording of mBOs placed in the triMEA platform. a) Electrochemical characterization of sMEA electrodes. i) Box plot of electrode impedance magnitude at 1 kHz across left (n = 30), center (n = 32), and right (n = 31) wells. ii) Bode plots of impedance magnitude (top) and phase (bottom) as a function of frequency for all three wells. b) Placement of mBOs in the triMEA platform. i) Optical images of mBOs seated in the left, center, and right wells. ii) Schematics of organoid positioning relative to the sMEA electrodes in each well. c) Representative extracellular recordings from five channels under continuous perfusion over 300 s across all three wells. d) Raster plots of detected spike events under perfusion across all three wells, color-coded by spike amplitude. e) Representative spike waveforms from channel 13 (left well) and channel 14 (center well). f) MAD noise comparison between static and flow conditions across left, center, and right wells. A statistically significant increase in noise floor was observed under perfusion at all three positions (Mann–Whitney U test with Bonferroni correction, $^{***}p < 0.001$, n = 16 channels per well per condition).

signal acquisition on the millifluidic triMEA organoid platform.

3.6 Simultaneous electrophysiology

The electrochemical properties of the sMEA were characterized prior to organoid loading by electrochemical impedance spectroscopy in PBS across a frequency range of 10 Hz to 1 MHz. Median impedance values of microelectrodes across the three wells remained within a narrow 2–7 kΩ range, which is sufficiently low to detect individual neuronal spikes (Fig. 5a) and is also consistent with previously reported low-impedance neural recording electrodes.[21,48–50] The conductive PEDOT : PSS layer increased the effective surface area of each electrode, which reduced the electrode-electrolyte interface impedance relative to bare gold. Impedance values measured following continuous perfusion at 0.1 mL min^{-1} remained within the same range, which indicated that the PEDOT : PSS coating was mechanically stable under flow conditions and that the electrode-electrolyte interface was not disrupted by continuous media perfusion.

Midbrain organoids were transferred to the three wells (Fig. 5b), and electrophysiological recordings were acquired over a 300 s window under continuous perfusion (Fig. 5c). Neural activities were detected in the left and center wells, with approximately 70 and 50 spikes recorded over the acquisition window, respectively, with amplitudes of approximately 40 to 50 μV. No detectable activities above the detection threshold were observed in the right well during the acquisition window. Raster plots revealed spontaneous bursting activities in the left well across channel 13, with sparser spiking detected in the center well, consistent with early-stage network activity characteristic of developing midbrain organoids[51,52] (Fig. 5d). The observed differences in spontaneous activities across wells reflect known organoid-to-organoid variability in electrophysiological maturation, as organoids derived from the same stem cell line under identical culture conditions do not necessarily reach the same functional state simultaneously, given that network formation and emergence of spontaneous activities are inherently stochastic processes.[10,53–55] Critically, because all three wells were recorded simultaneously under identical mechanical, biochemical, and electrical conditions, these differences are likely attributable to the biological state of each organoid rather than to technical inconsistencies in the recording environment, which would not be possible with sequential single-well recordings. To confirm that the recorded activity originated from the organoid rather than from the recording system or fluidic environment, recordings were acquired from a representative device across active channels before organoid loading. No spike events were detected under this condition, and clear spike activity emerged on channel 3 following organoid loading (Fig. S9a). The averaged trace from this channel before loading resembled the baseline noise, whereas the averaged trace after loading exhibited a sharp positive deflection followed by a negative recovery phase, characteristic of an extracellular action potential (Fig. S9b). These results were consistent with prior validation of the same sMEA and organoid type previously reported by our group.[21] Representative neuronal spike waveforms extracted from channel 13 of the left well and channel 14 of the center well exhibited the characteristic biphasic morphology of extracellular action potentials, with a sharp positive deflection followed by a negative recovery phase (Fig. 5e). To characterize the effect of continuous perfusion on recording quality, MAD noise was compared between static and perfused conditions across all three wells (Fig. 5f). A statistically significant increase in noise floor was observed under flow conditions, with MAD noise rising from approximately 2 μV under static conditions to approximately 2.5 to 3.2 μV under perfusion. Despite the slight increase, the absolute noise values remained sufficiently low to permit reliable detection of neuronal spikes.

The measured electrophysiological signals confirmed that the triMEA platform performed as designed, with matched perfusion, stable electrode impedance, and consistent recording baseline maintained throughout the acquisition window. Triplicate recordings within a single experiment eliminated the temporal variability inherent to the sequential single-well format, where differences in media composition, organoid age, and environmental conditions between experiments introduce confounding factors that are difficult to decouple from genuine biological differences in neural activities. The shared inlet architecture ensured that all three organoids were exposed to identical media composition and flow conditions simultaneously, which implies that the microenvironment at each recording site is governed by the same physical and biochemical parameters. The electrophysiological demonstration presented here represents a single proof-of-concept session intended to establish platform functionality under biologically relevant conditions, with systematic characterization across multiple organoid batches and maturation stages representing a future direction for our work.

4 Conclusion

In this work, we developed the triMEA millifluidic platform, which to our knowledge is the first device to integrate dedicated perfusion with electrophysiological recording *via* a three-dimensional, stretchable microelectrode array, across three independent parallel wells. The system combines controlled flow and shear stress distribution with simultaneous, multi-site signal acquisition under continuous perfusion. This integration enables a three-fold increase in electrophysiological throughput while preserving consistent culture conditions across wells. The platform can be adapted to applications ranging from longitudinal functional characterization to studies of organoid-to-organoid variability. Beyond these immediate uses, the triMEA system provides a foundation for next-generation organoid technologies that require integrated control of fluidic, electrical, and biological environments. This work demonstrates that microfluidics, electrophysiology, and biological interfacing can be successfully integrated within a single platform, and points toward future incorporation of complementary technologies such as optical interrogation and closed-loop

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systems. The perfusion-based architecture further opens the door to a broader range of compound delivery strategies, from passive to active on-site delivery through emerging tools, such as digital microfluidics,[56,57] acoustic actuators,[58,59] and microrobot-based[60,61] approaches.

Author contributions

E. C., H. J. L., K. K. and Y. Kim conceived the study. K. K. and Y. Kim contributed to the methodology of the electrophysiology measurements and the fabrication of stretchable microelectrode arrays. E. C. designed the platform and experiments, fabricated the stretchable microelectrode arrays and millifluidic device, implemented the analysis and visualization software, conducted the millifluidic and electrophysiology experiments, and performed the formal data analysis. J. A. G. contributed to the data analysis. Y. K. generated organoids from stem cells, performed validation and phase-contrast imaging, and provided culture medium. Y. L. assisted with organoid generation and culture medium preparation. E. C. and H. J. maintained the organoids. H. J. assisted with experiments, figure preparation, and the drafting of selected sections. K. K., M. O. L. and H. J. L. supervised the project. M. O. L. and H. J. L. managed project administration and secured funding. E. C. wrote the manuscript, and H. J. L. revised it, with all authors commenting on and approving the final version.

Conflicts of interest

There are no conflicts to declare.

Data availability

The data supporting the findings of this study are available within the article and its supplementary information (SI).

Supplementary information: contains the fabrication process of the stretchable microelectrode array (Fig. S1), the millifluidic mold design (Fig. S2), the assembled device with the optical adhesive seal (Fig. S3), the flow distributions of the initial series and parallel channel layouts (Fig. S4), the effects of side channel width section 2 and channel height on shear stress (Fig. S5), the robustness of the optimized flow distribution to well geometry, organoid size, and the presence of the sMEA (Fig. S6), the shear stress distribution on the optimized geometry with the sMEA included (Fig. S7), the oxygen transport model under static and perfused conditions (Fig. S8), the recordings acquired before and after organoid loading (Fig. S9), and reported ranges of physiologically tolerable shear stress (Table S1). See DOI: https://doi.org/10.1039/d6lc00391e.

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